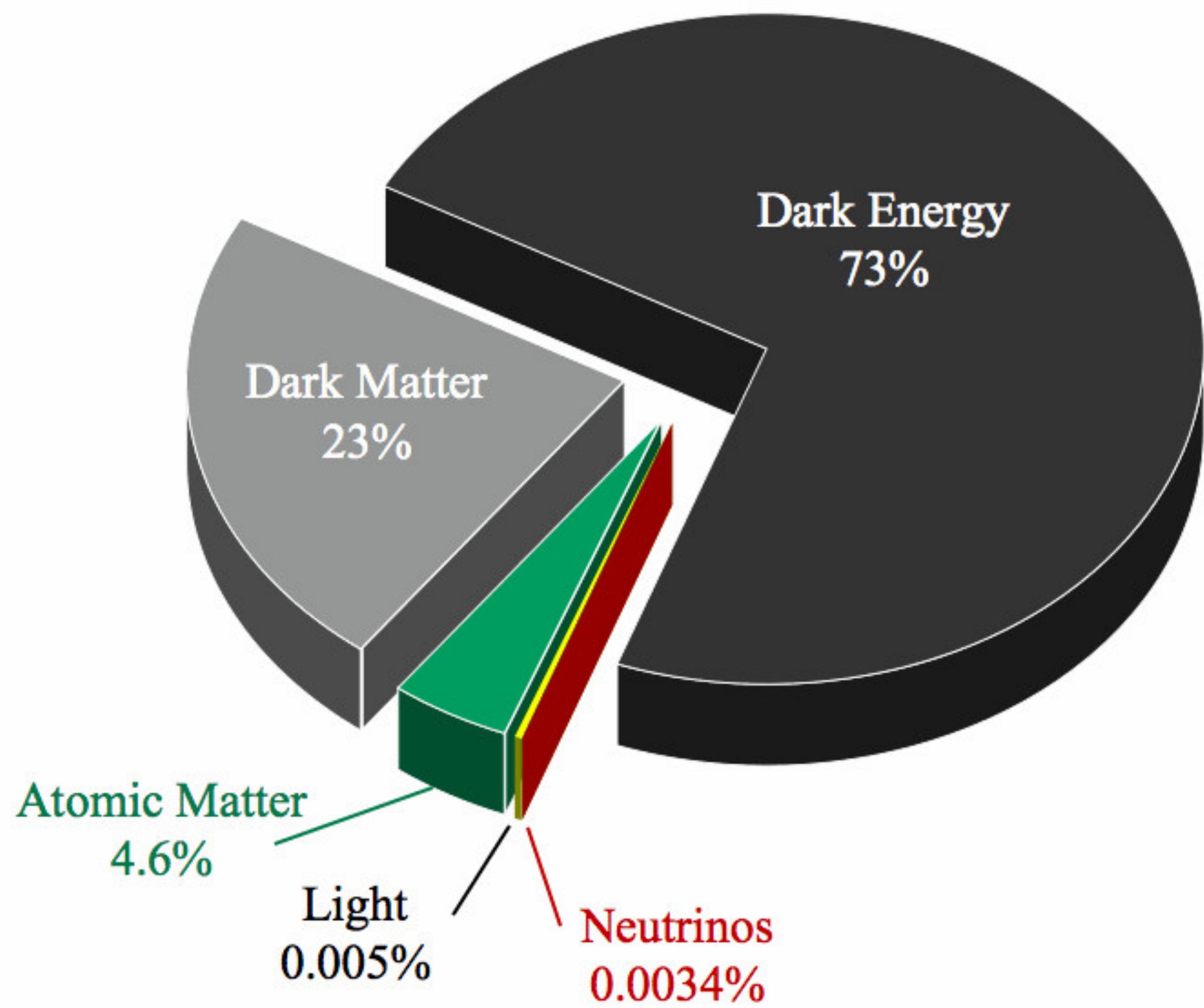


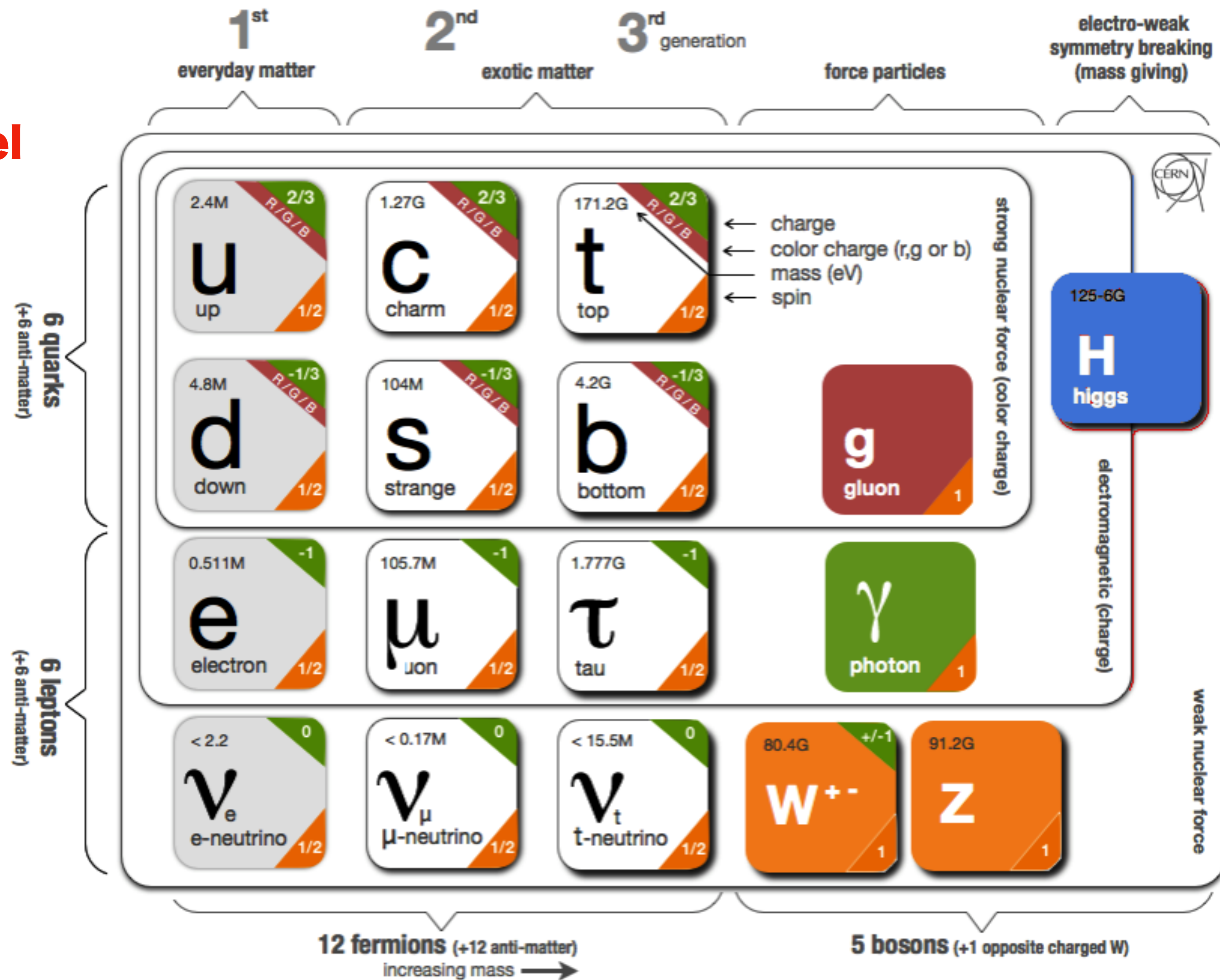
Introduction to Particle Physics

What is the world made of?

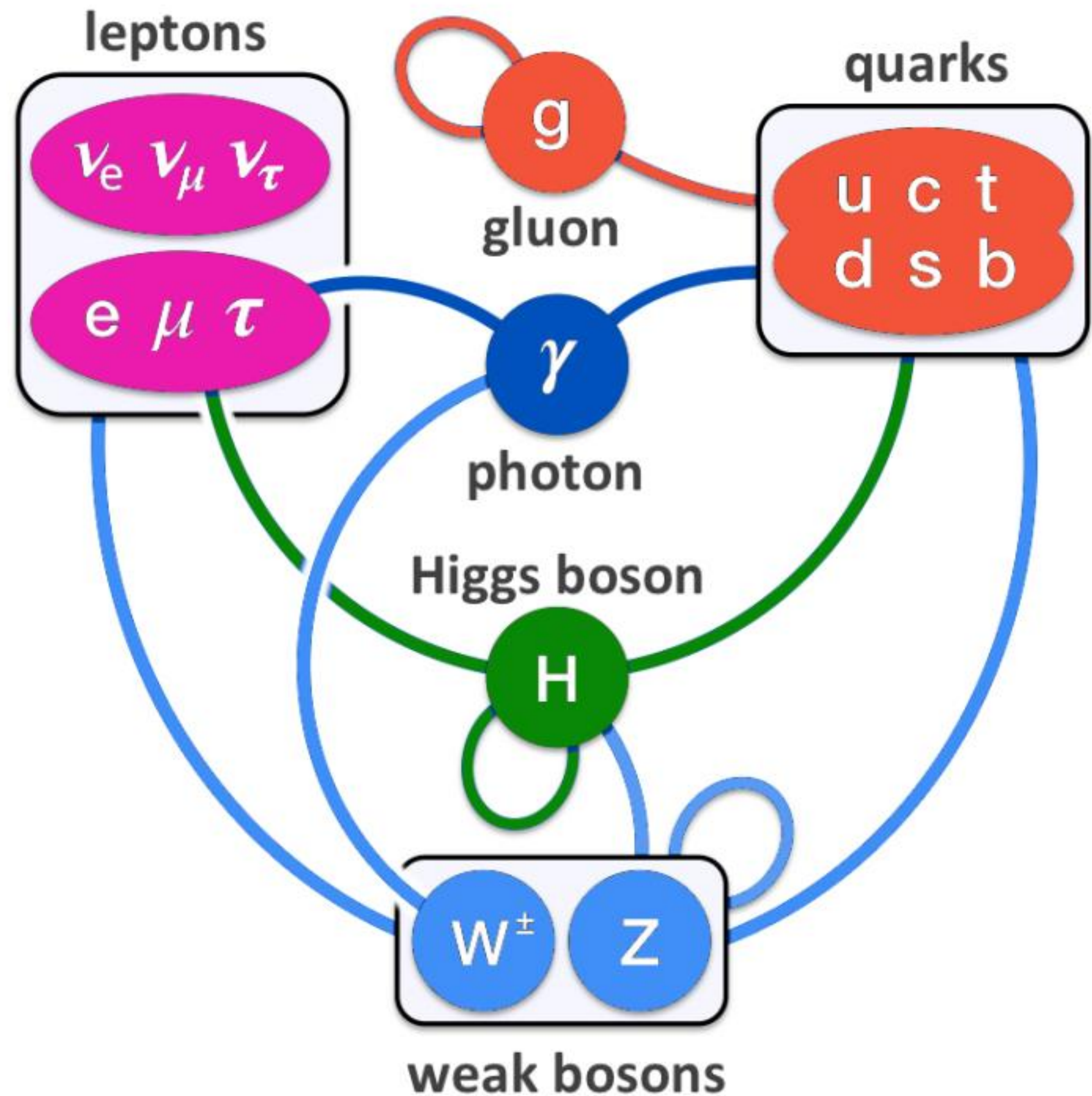


What is the visible world made of?

The Standard Model of Particle Physics



The Standard Model of Particle Physics



The Standard Model of Particle Physics



A good set of units

SI units are NOT useful. Too big!

Make life simple by setting $c = \hbar = 1$

$$1\text{ eV} = 1.6 \times 10^{-19} J$$

$$\begin{aligned} 1\text{ TeV} &= 10^3\text{ GeV} = 10^6\text{ MeV} = 10^9\text{ KeV} = 10^{12}\text{ eV} \\ 1\text{ fermi} &\equiv 1\text{ F} = 10^{-13}\text{ cm} = 5.07\text{ GeV}^{-1} \\ (1\text{ F})^2 &= 10\text{ mb} = 10^4\text{ }\mu\text{b} = 10^7\text{ nb} = 10^{10}\text{ pb} \\ (1\text{ GeV})^{-2} &= 0.389\text{ mb} \end{aligned}$$

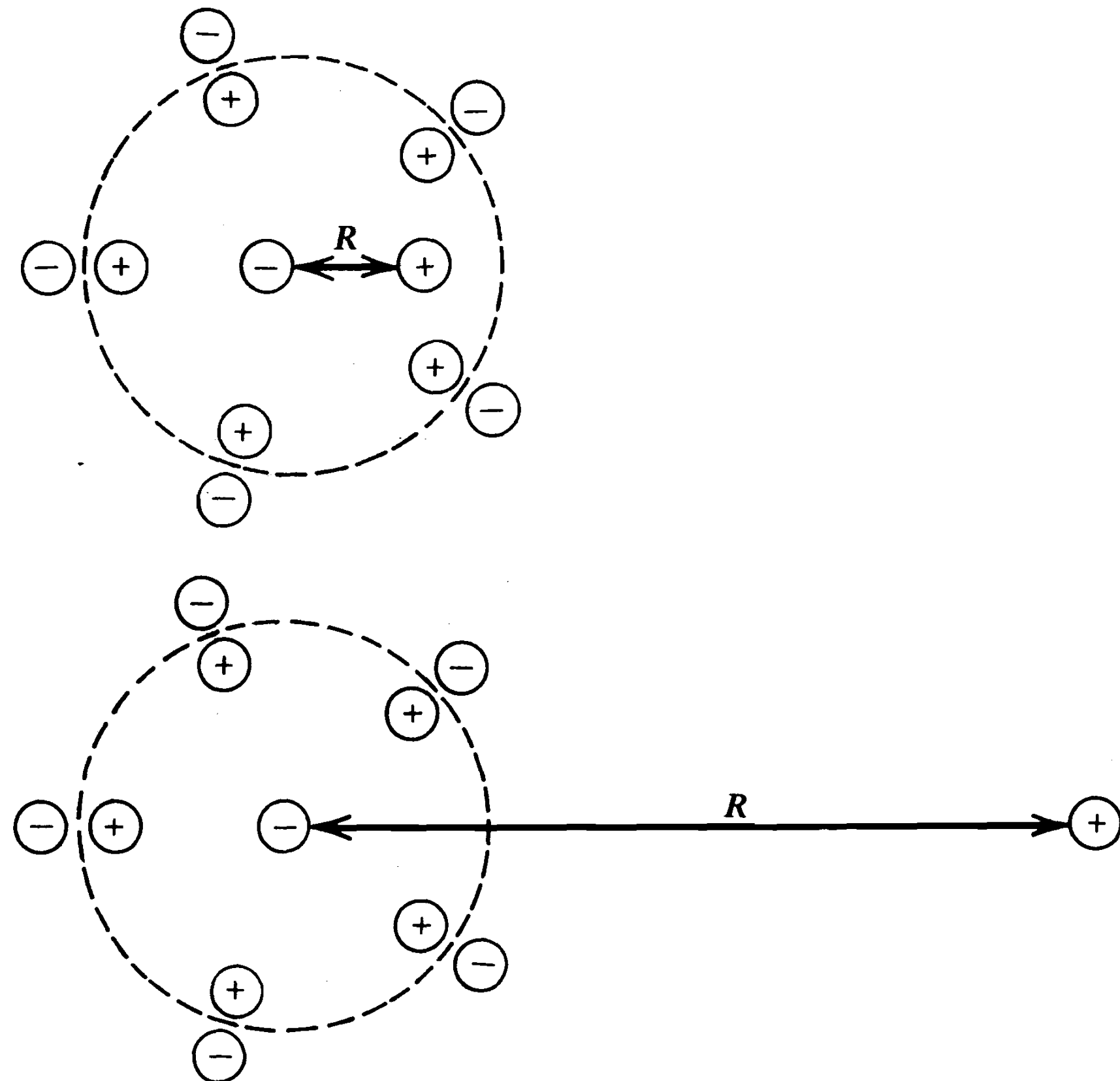
Actually, particle theorists are lazy (or clever, depending on your point of view)—they seldom include the c 's and $\hbar$'s ($\hbar \equiv h/2\pi$) in their formulas. You're just supposed to fit them in for yourself at the end, to make the dimensions come out right. As they say in the business, "set $c = \hbar = 1$." This amounts to working in units such that time is measured in centimeters and mass and energy in inverse centimeters; the unit of time is the time it takes light to travel 1 centimeter, and the unit of energy is the energy of a photon whose wavelength is 2π centimeters.

Conversion Factor	$\hbar = c = 1$ Units	Actual Dimension
$1\text{ kg} = 5.61 \times 10^{26}\text{ GeV}$	GeV	$\frac{\text{GeV}}{c^2}$
$1\text{ m} = 5.07 \times 10^{15}\text{ GeV}^{-1}$	GeV^{-1}	$\frac{\hbar c}{\text{GeV}}$
$1\text{ sec} = 1.52 \times 10^{24}\text{ GeV}^{-1}$	GeV^{-1}	$\frac{\hbar}{\text{GeV}}$
$e = \sqrt{4\pi\alpha}$	—	$(\hbar c)^{1/2}$

By choosing units with $\hbar = c = 1$, it becomes unnecessary to write $\hbar$ and c explicitly in the formulas, thus saving a lot of time and trouble. We can always use dimensional analysis to work out unambiguously where the $\hbar$'s and c 's enter any formula. Hence, with a slight but permissible laziness, it is customary to speak of mass (m), momentum (mc), and energy (mc^2) all in terms of GeV, and to measure length ($\hbar/mc$) and time ($\hbar/mc^2$) in units of GeV^{-1} .

A good set of units

The charge of an electron



So far, we have not considered the elementary charge e , which measures how strongly electrons, say, interact electromagnetically with each other. To obtain a dimensionless measure of the strength of this interaction, we compare the electrostatic energy of repulsion between two electrons one natural unit of length apart with the rest mass energy of an electron:

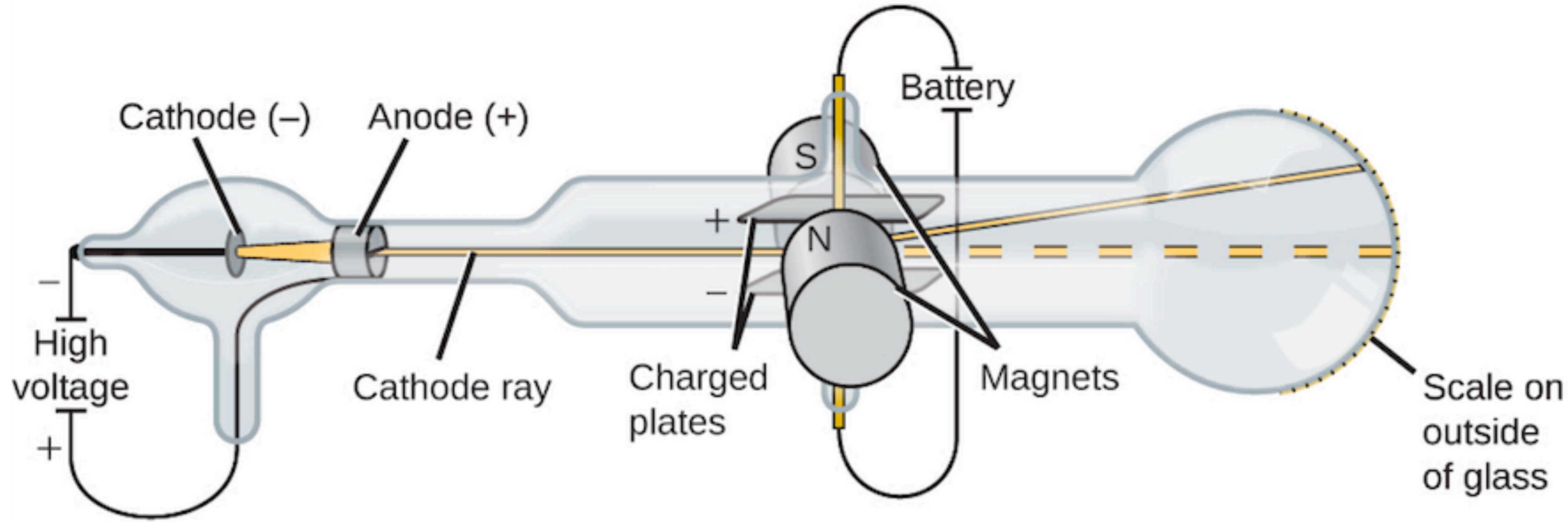
$$\alpha = \frac{1}{4\pi} \frac{e^2}{(\hbar/mc)} / mc^2 = \frac{e^2}{4\pi\hbar c} \approx \frac{1}{137}.$$

we have adopted the rationalized Heaviside–Lorentz system of electromagnetic units. That is, the 4π factors appear in the force equations rather than in the Maxwell equations, and ϵ_0 is set equal to unity. This choice, which is conventional in particle physics, reduces Maxwell's equations to their simplest possible form. The value of α is, of course, the same in all systems of units, but the numerical value of e is different.

For historical reasons, α is known as the fine structure constant. Unfortunately, this name conveys a false impression. We have seen that the charge of an electron is not strictly constant but varies with distance because of quantum effects; hence α must be regarded as a variable, too. The value $\frac{1}{137}$ is the asymptotic value of α

History Lesson

Electron discovery, 1897

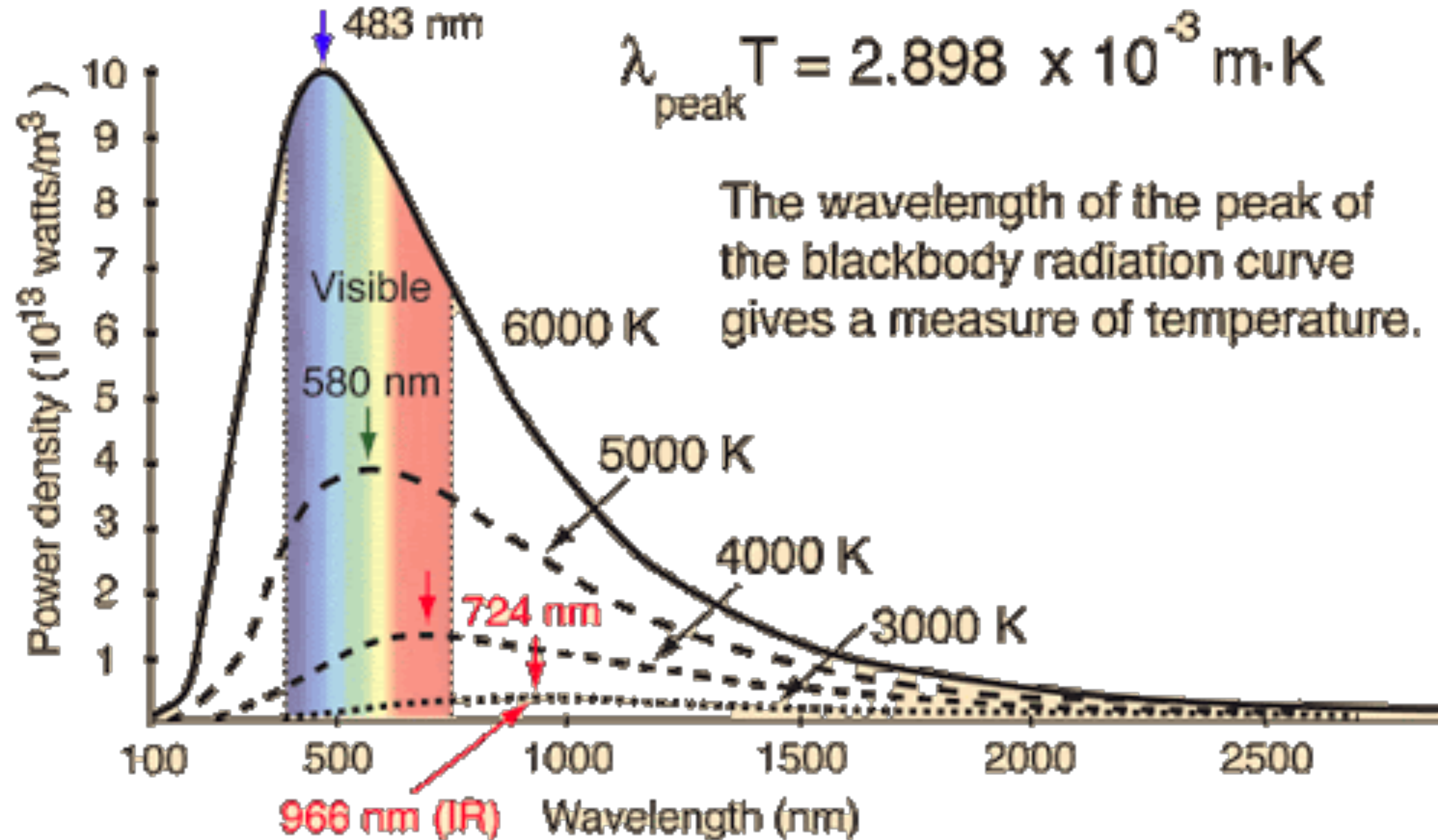


Joseph John Thomson

- Radioactivity was discovered in 1896 by **Henri Becquerel**, while working with phosphorescent materials.
- In 1898, **Ernest Rutherford** at Cavendish Laboratory distinguished two types of radioactivity, alpha rays and beta rays, which differed in their ability to penetrate, or travel into, ordinary objects or gases.

History Lesson

Planck backbody spectrum, 1900

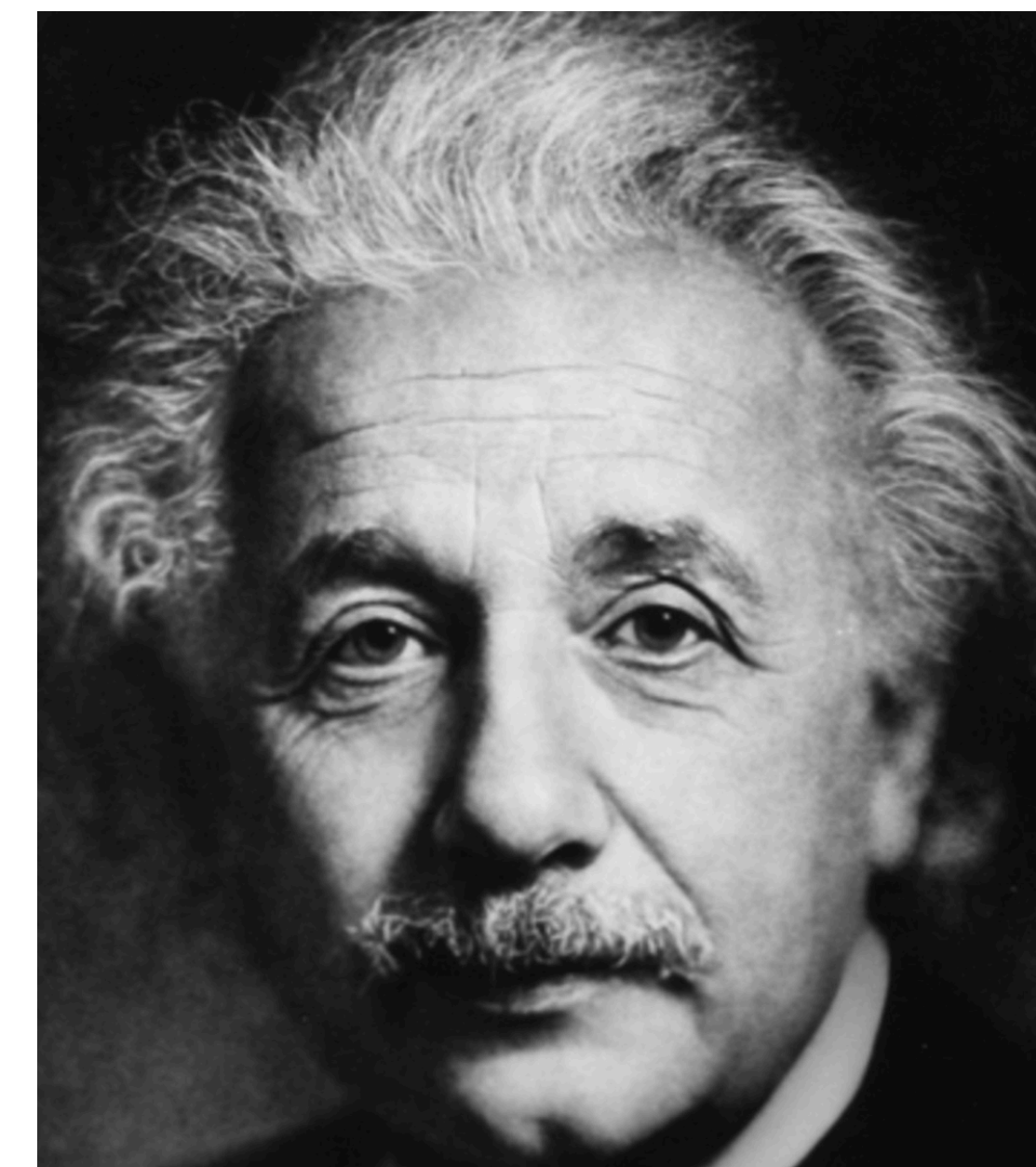
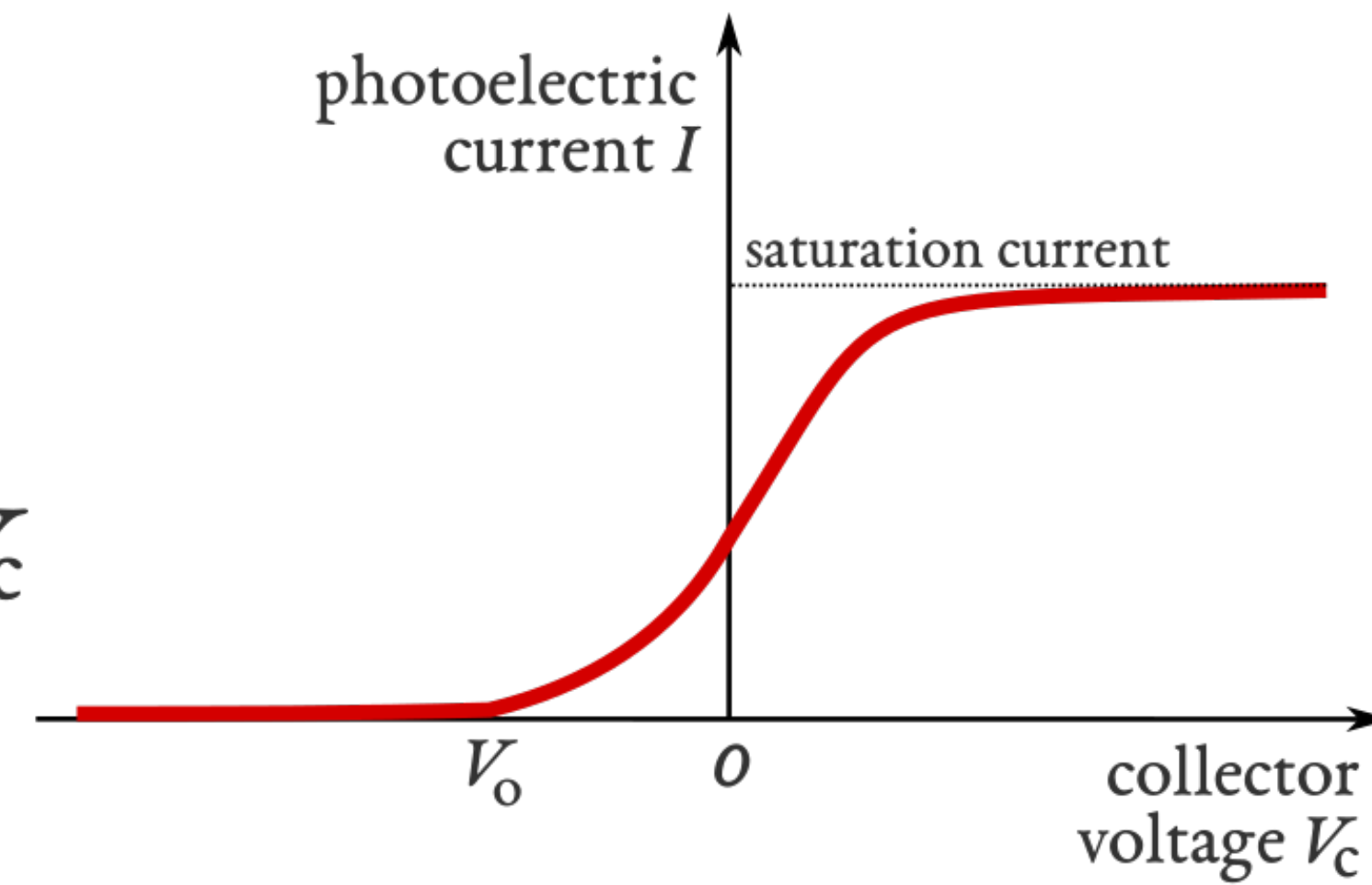
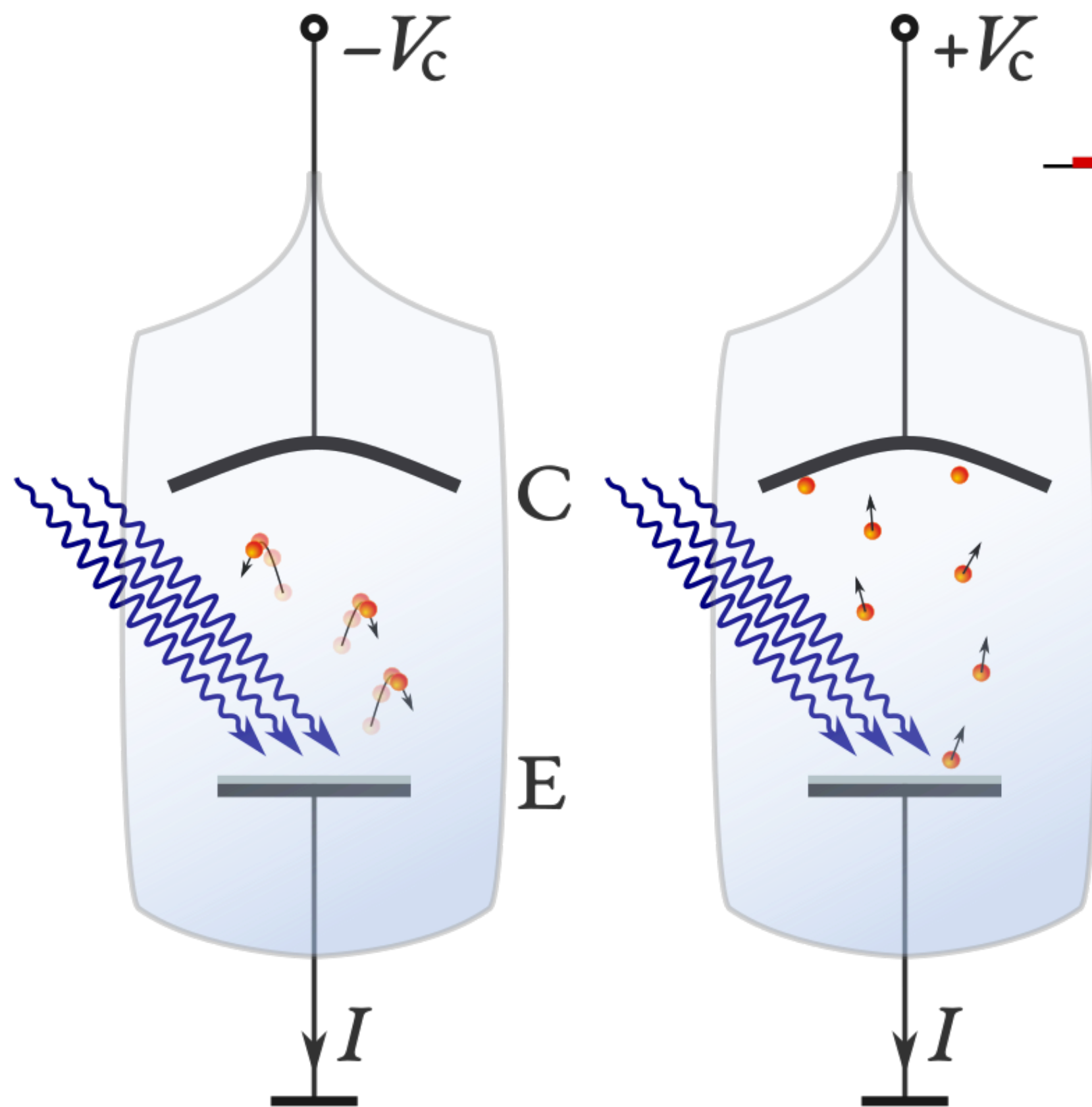


Max Karl Ernst Ludwig Planck

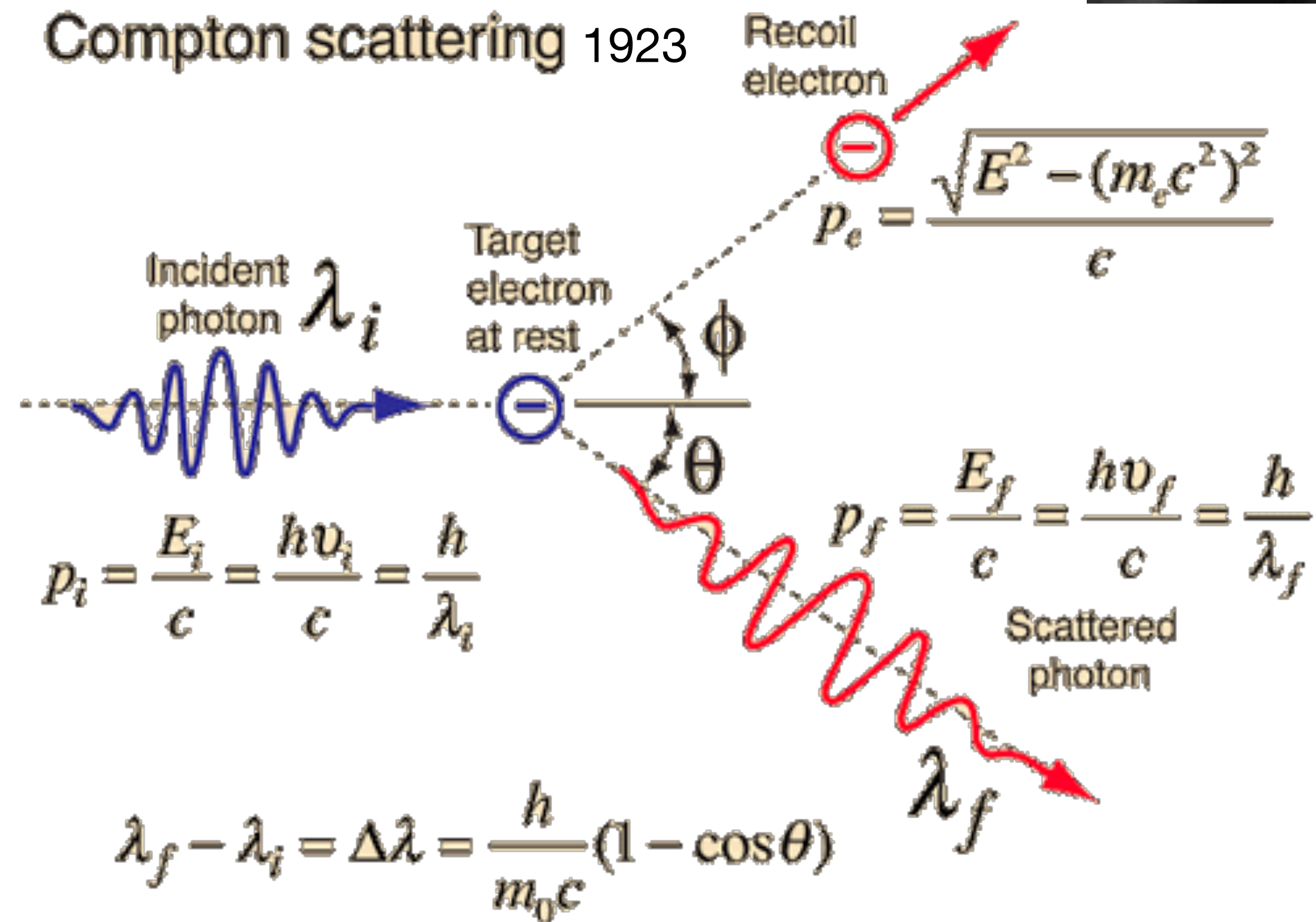
- **Paul Villard** discovered gamma rays.
- Beta rays were shown to be electrons by **Walter Kaufmann** in 1902

History Lesson

Photoelectric effect, 1905



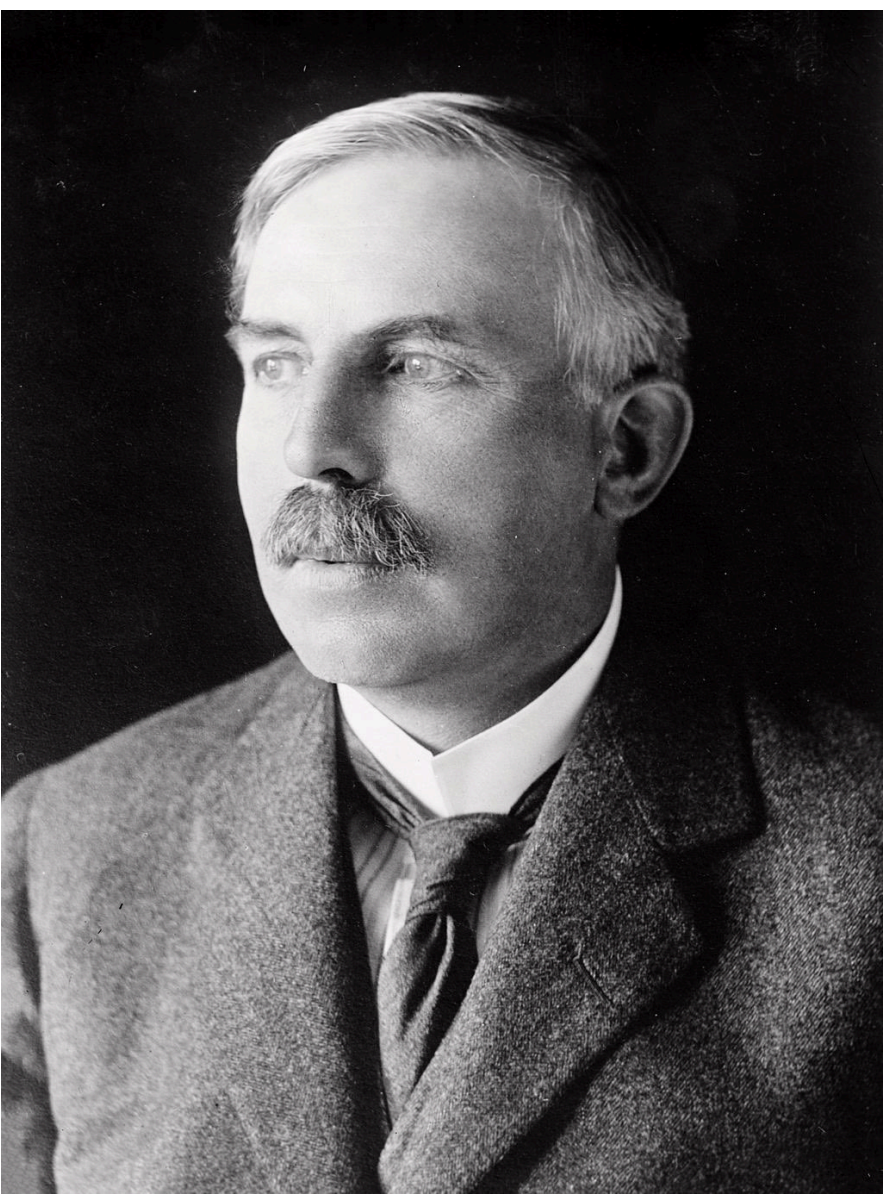
Compton scattering 1923



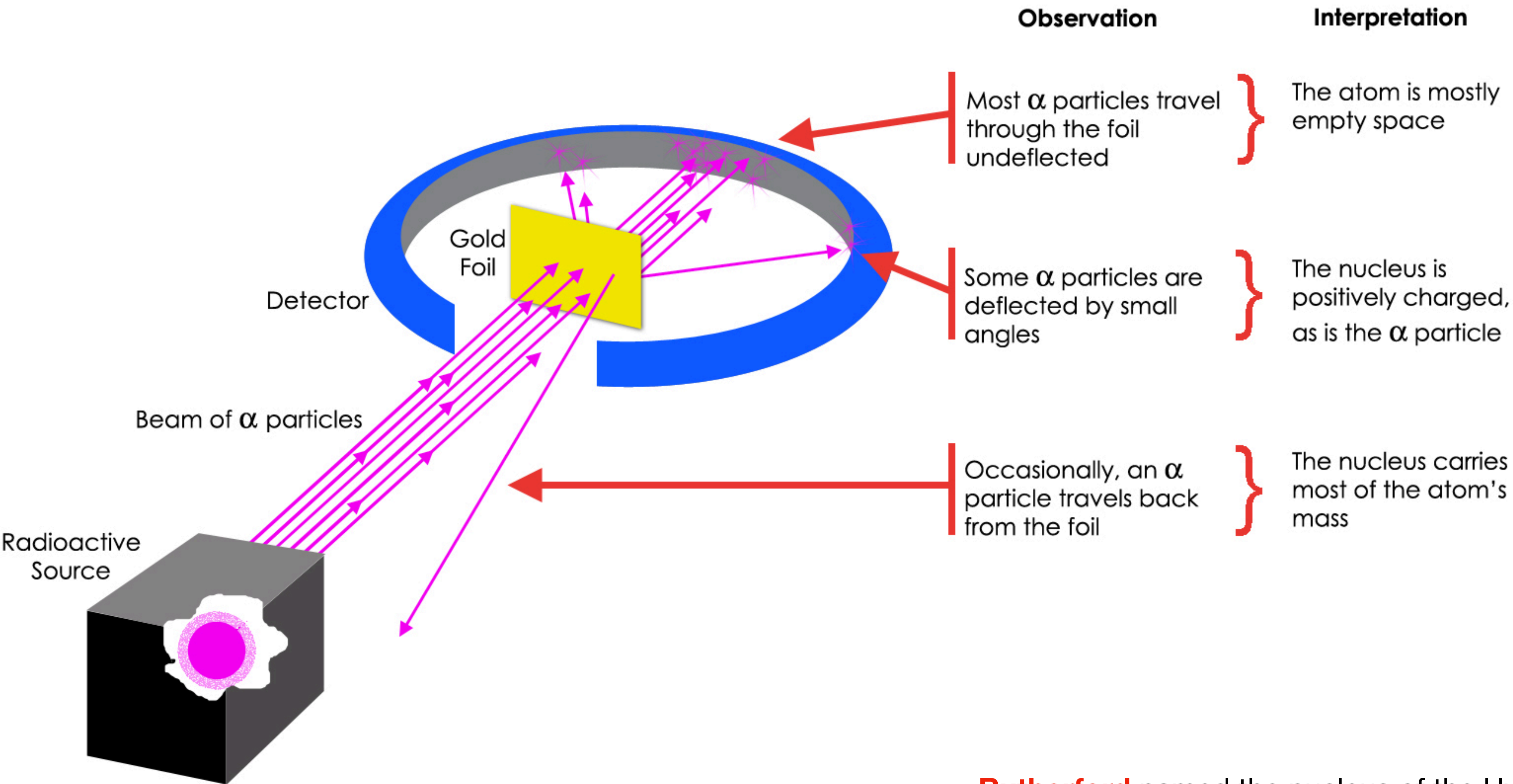
- Alpha rays were shown to be helium ions by **Rutherford** and **Thomas Royds** in 1907

History Lesson

Rutherford's experiment, 1911



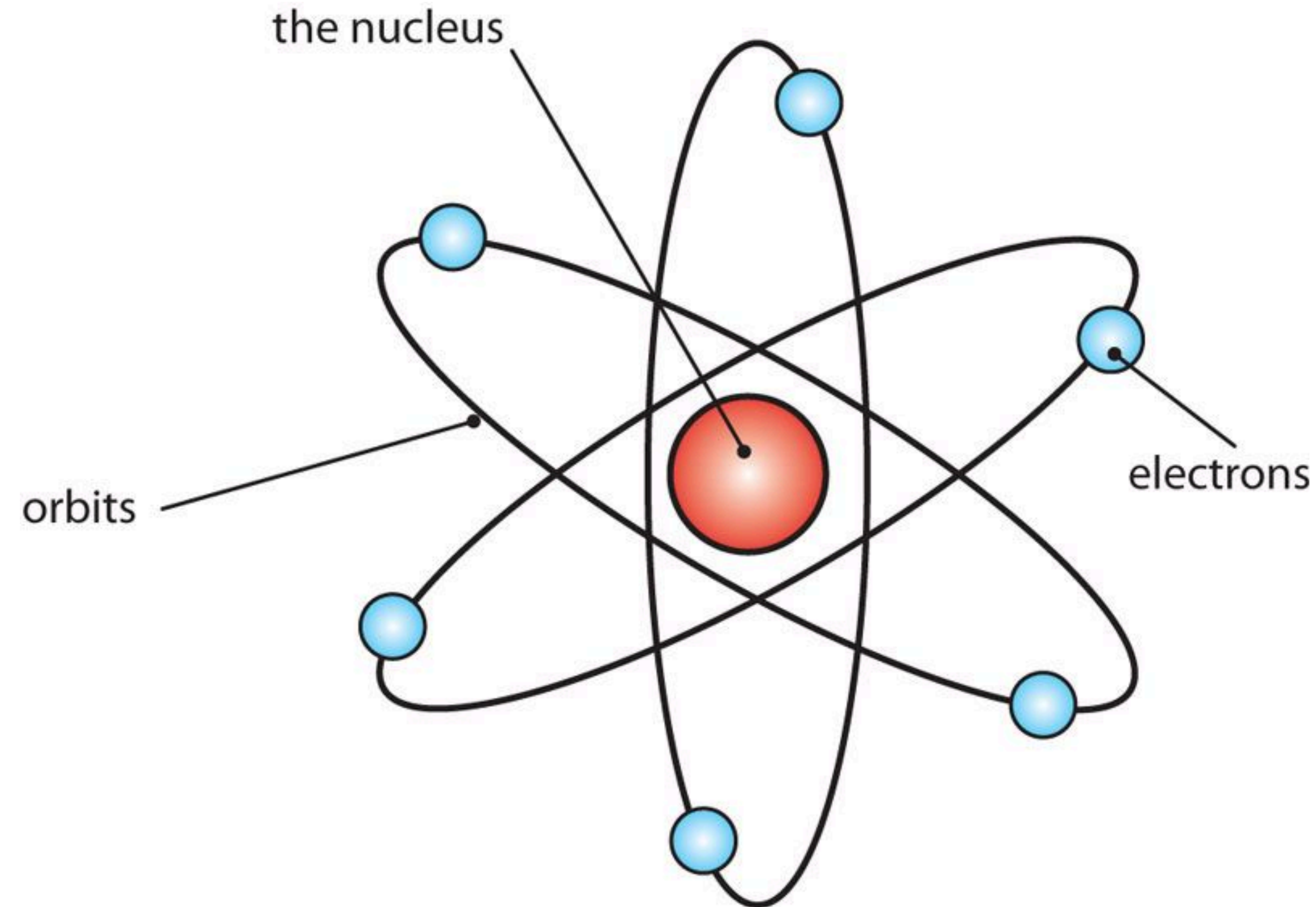
Ernest Rutherford



Rutherford named the nucleus of the Hydrogen atom as “Proton”

History Lesson

Bohr's model of atom, 1914

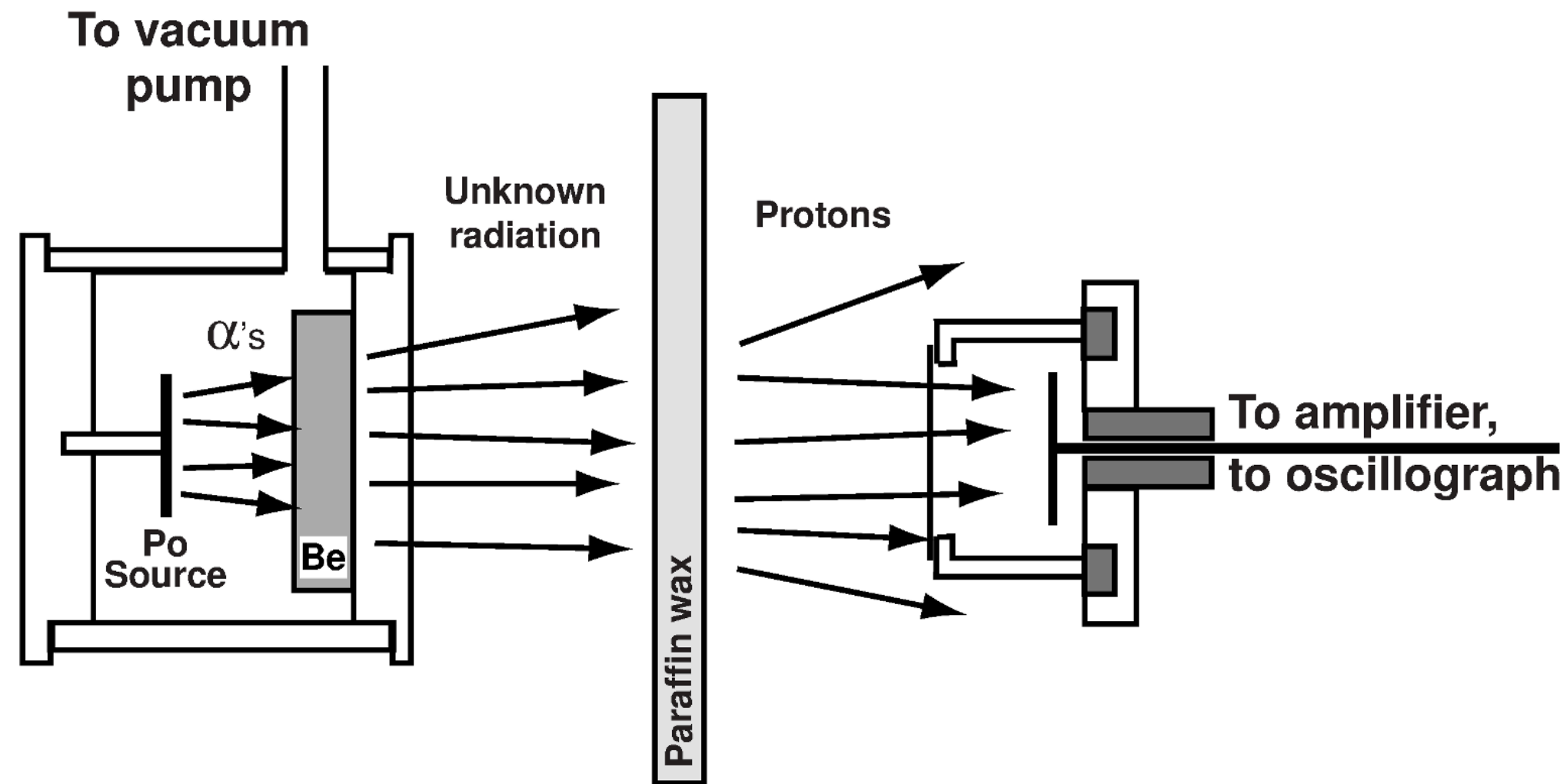


Niels Henrik David Bohr

- In 1913, **Frederick Soddy** and **Kazimierz Fajans** independently found that an element undergoing alpha decay will produce an element two places to the left in the periodic system and an element undergoing beta decay will produce an element one place to the right in the periodic system. Also, those radioelements that reside in the same places in the periodic system are chemically identical. Soddy called these chemically identical elements **isotopes**.
- 1914 - Gamma rays were shown to be electromagnetic radiation by **Rutherford** and **Edward Andrade**.

History Lesson

Discovery of neutron, 1932



James Chadwick

- In 1930, **Walther Bothe** and **Herbert Becker** found that if the energetic alpha particles emitted from polonium fell on certain light elements, an unusually penetrating radiation was produced. Beryllium produced the most intense radiation. Polonium is highly radioactive, producing energetic alpha radiation. The observed penetrating radiation was not influenced by an electric field, however, so it was thought to be gamma radiation. The radiation was more penetrating than any gamma rays known, and the details of experimental results were difficult to interpret.
- In 1932, **Irène Joliot-Curie** and **Frédéric Joliot** in Paris showed that if this unknown radiation fell on paraffin wax, or any other hydrogen-containing compound, it ejected protons of very high energy (5 MeV). This observation was not in itself inconsistent with the assumed gamma ray nature of the new radiation, but that interpretation (Compton scattering) had a logical problem. From energy and momentum considerations, a gamma ray would have to have impossibly high energy (50 MeV) to scatter a massive proton.
- Assisted by **Norman Feather**, **James Chadwick** at the Cavendish Laboratory quickly performed a series of experiments showing that the gamma ray hypothesis was untenable.
- Chadwick measured the range of these protons, and also measured how the new radiation impacted the atoms of various gases. He found that the new radiation consisted of not gamma rays, but uncharged particles with about the same mass as the proton. These particles were **neutrons**.

What Binds the Nucleus?

The first significant theory of the strong force was proposed by Yukawa in 1934.

Yukawa assumed that the proton and neutron are attracted to one another by some sort of *field*, just as the electron is attracted to the nucleus by an electric field and the moon to the earth by a gravitational field. This field should properly be quantized, and Yukawa asked the question: what must be the properties of its *quantum* – the particle (analogous to the photon) whose exchange would account for the known features of the strong force? For example, the short range of the force indicated that the mediator would be rather heavy; Yukawa calculated that its mass should be nearly 300 times that of the electron, or about a sixth the mass of a proton

Because it fell between the electron and the proton, Yukawa's particle came to be known as the *meson* (meaning 'middle-weight'). In the same spirit, the electron is called a *lepton* ('light-weight'), whereas the proton and neutron are *baryons* ('heavy-weight'). Now, Yukawa knew that no such particle had ever been observed in the laboratory, and he therefore assumed his theory was wrong. But at that time a number of systematic studies of cosmic rays were in progress, and by 1937 two separate groups (Anderson and Neddermeyer on the West Coast, and Street and Stevenson on the East) had identified particles matching Yukawa's description. Indeed, the cosmic rays with which you are being bombarded every few seconds as you read this consist primarily of just such middle-weight particles.



Hideki Yukawa

What Binds the Nucleus?

For a while everything seemed to be in order. But as more detailed studies of the cosmic ray particles were undertaken, disturbing discrepancies began to appear. They had the wrong lifetime and they seemed to be significantly lighter than Yukawa had predicted; worse still, different mass measurements were not consistent with one another. In 1946 (after a period in which physicists were engaged in a less savory business) decisive experiments were carried out in Rome demonstrating that the cosmic ray particles interacted very weakly with atomic nuclei [5]. If this was really Yukawa's meson, the transmitter of the strong force, the interaction should have been dramatic. The puzzle was finally resolved in 1947, when Powell and his coworkers at Bristol [6] discovered that there are actually two middle-weight particles in cosmic rays, which they called π (or 'pion') and μ (or 'muon'). (Marshak reached the same conclusion simultaneously, on theoretical grounds [7].) The true Yukawa meson is the π ; it is produced copiously in the upper atmosphere, but ordinarily disintegrates long before reaching the ground

Powell's group exposed their photographic emulsions on mountain tops

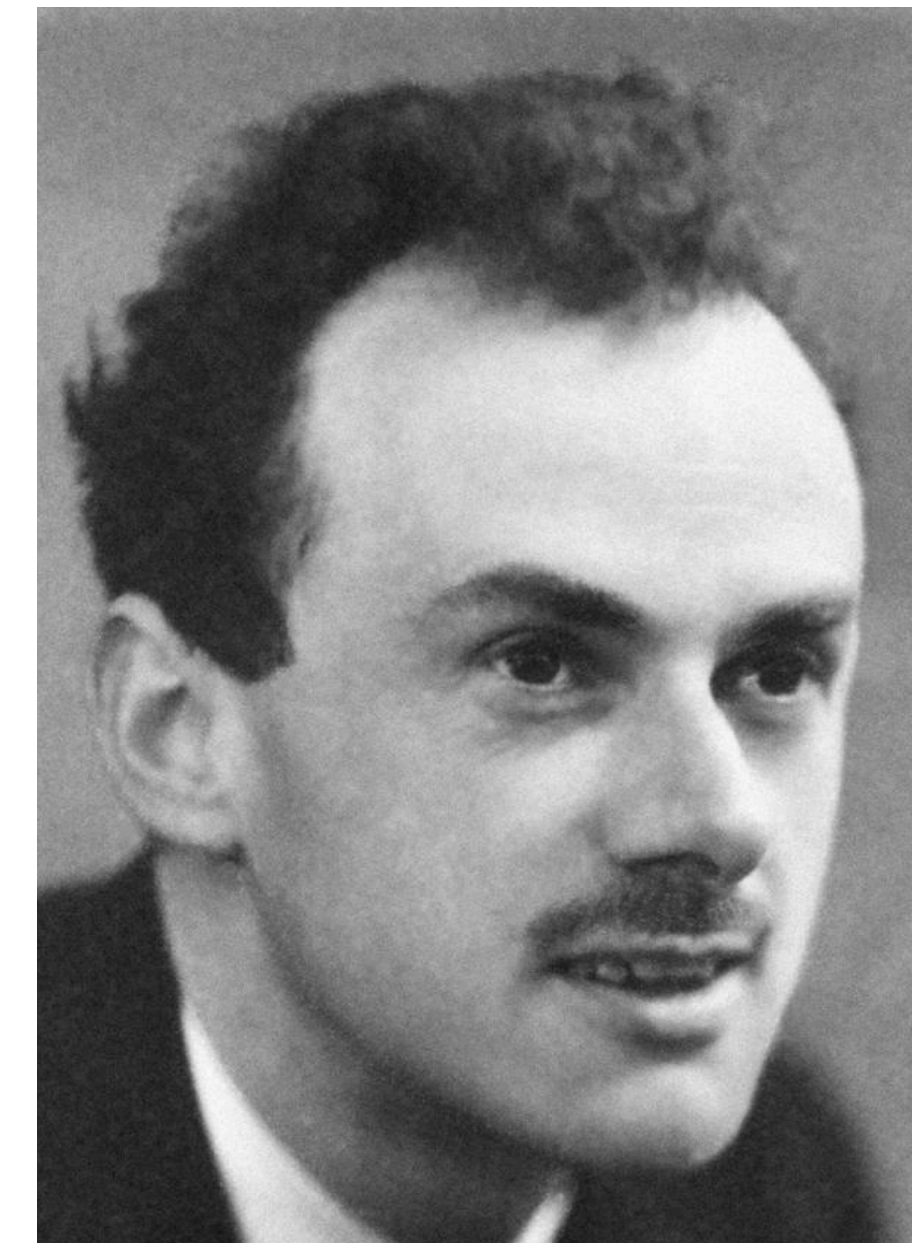
One of the decay products is the lighter (and longer lived) μ , and it is primarily muons that one observes at sea level. In the search for Yukawa's meson, then, the muon was simply an impostor, having nothing whatever to do with the strong interactions. In fact, it behaves in every way like a heavier version of the electron and properly belongs in the *lepton* family (though some people to this day call it the 'mu-meson' by force of habit).



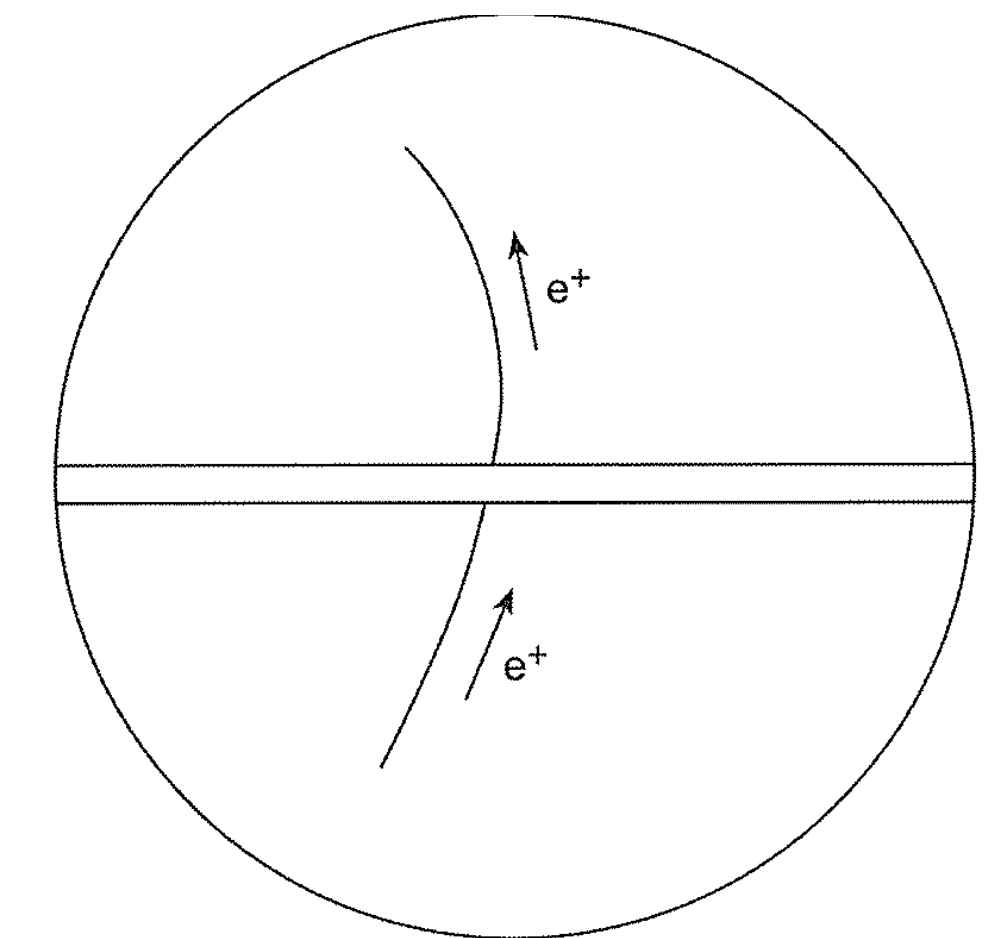
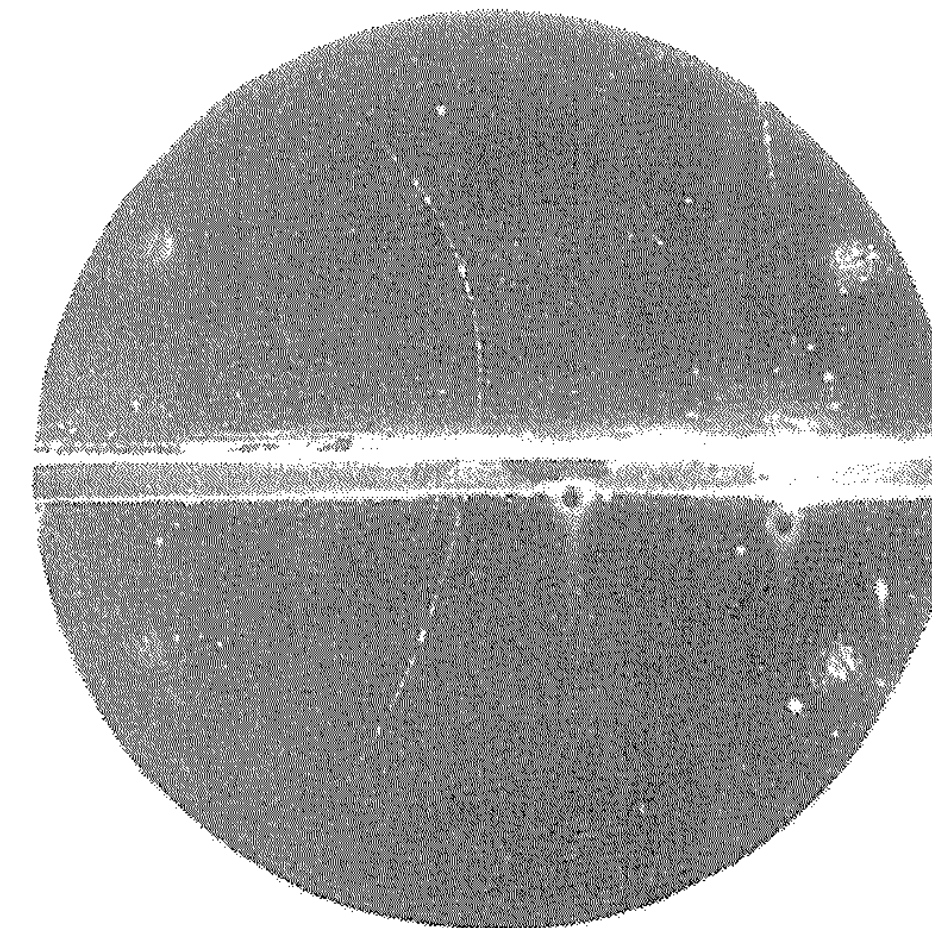
Hideki Yukawa

Dirac Equation - Antiparticles

The first major achievement was Dirac's discovery, in 1927, of the equation that bears his name. The Dirac equation was supposed to describe free electrons with energy given by the relativistic formula $E^2 - \mathbf{p}^2 c^2 = m^2 c^4$. But it had a very troubling feature: for every positive-energy solution ($E = +\sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$) it admitted a corresponding solution with *negative* energy ($E = -\sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$). This meant that, given the natural tendency of every system to evolve in the direction of lower energy, the electron should 'runaway' to increasingly negative states, radiating off an infinite amount of energy in the process. To rescue his equation, Dirac proposed a resolution that made up in brilliance for what it lacked in plausibility: he postulated that the negative-energy states are all filled by an infinite 'sea' of electrons. Because this sea is always there, and perfectly uniform, it exerts no net force on anything, and we are not normally aware of it. Dirac then invoked the Pauli exclusion principle (which says that no two electrons can occupy the same state), to 'explain' why the electrons we *do* observe are confined to the positive-energy states. But if this is true, then what happens when we impart to one of the electrons in the 'sea' an energy sufficient to knock it into a positive-energy state? The *absence* of the 'expected' electron in the sea would be interpreted as a net positive charge in that location, and the absence of its expected negative energy would be seen as a net positive energy. Thus a 'hole in the sea' would function as an ordinary particle with *positive* energy and *positive* charge. Dirac at first hoped that these holes might be *protons*, but it was soon apparent that they had to carry the same mass as the electron itself – 2000 times too light to be a proton. No such particle was known at the time, and Dirac's theory appeared to be in trouble. What may have seemed a fatal defect in 1930, however, turned into a spectacular triumph in late 1931, with Anderson's discovery of the *positron*



Paul Dirac



Fun with Antiparticles

There is a general principle in particle physics that goes under the name of *crossing symmetry*. Suppose that a reaction of the form

$$A + B \rightarrow C + D$$

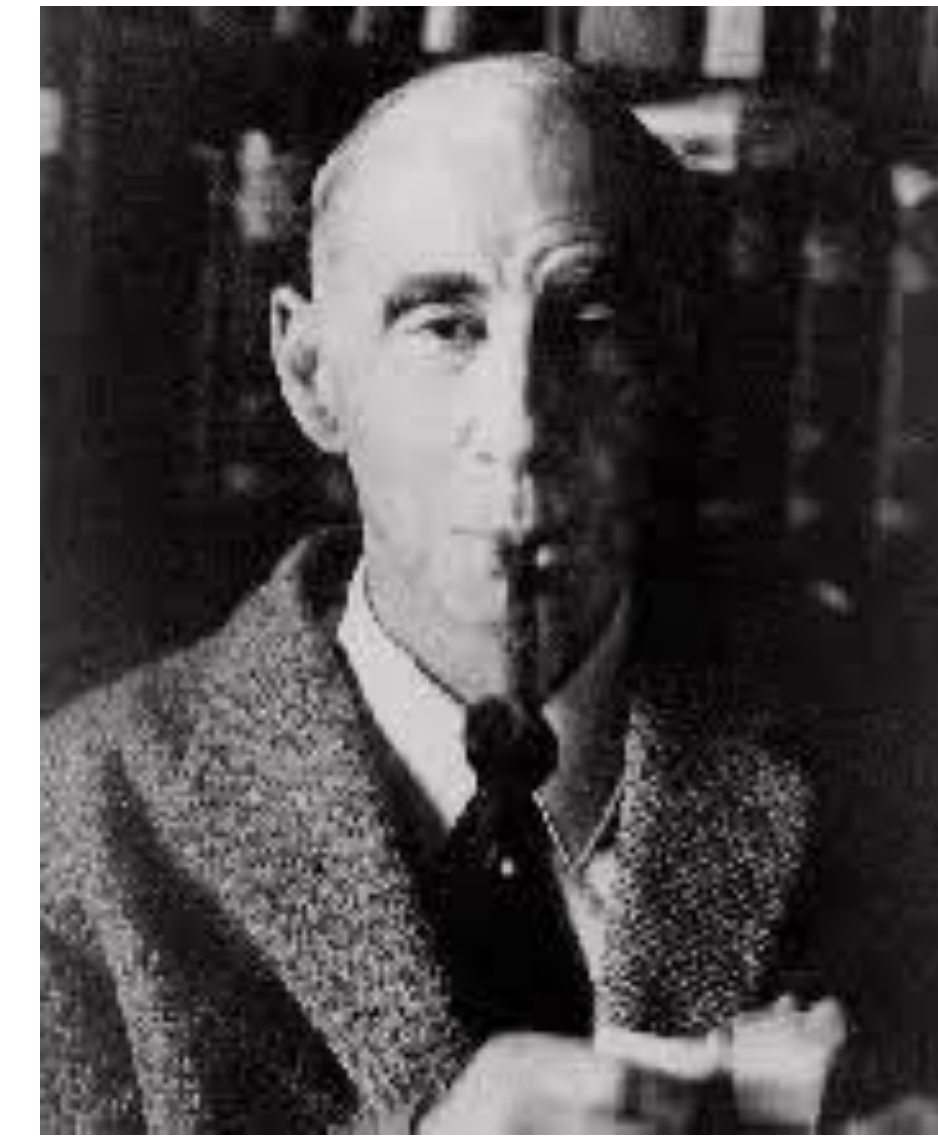
is known to occur. Any of these particles can be ‘crossed’ over to the other side of the equation, provided it is turned into its antiparticle, and the resulting interaction will also be allowed. For example,

$$A \rightarrow \bar{B} + C + D$$

$$A + \bar{C} \rightarrow \bar{B} + D$$

$$\bar{C} + \bar{D} \rightarrow \bar{A} + \bar{B}$$

In addition, the *reverse* reaction occurs: $C + D \rightarrow A + B$, but technically this derives from the principle of *detailed balance*, rather than from crossing symmetry. Indeed, as we shall see, the *calculations* involved in these various reactions are practically identical. We might almost regard them as different manifestations of the same fundamental process. However, there is one important *caveat* in all this: conservation of energy may veto a reaction that is otherwise permissible. For example, if A weighs less than the sum of B , C , and D , then the decay $A \rightarrow \bar{B} + C + D$ cannot occur; similarly, if A and C are light, whereas B and D are heavy, then the reaction $A + \bar{C} \rightarrow \bar{B} + D$ will not take place unless the initial kinetic energy exceeds a certain ‘threshold’ value. So perhaps I should say that the crossed (or reversed) reaction is *dynamically* permissible, but it may or may not be *kinematically* allowed. The power and beauty of crossing symmetry can scarcely be exaggerated. It tells us, for instance, that Compton scattering



Ernst Carl Gerlach Stueckelberg



Richard Feynman

In the Feynman–Stueckelberg formulation, the negative-energy solutions are re-expressed as *positive-energy* states of a *different particle* (the positron); the electron and positron appear on an equal footing, and there is no need for Dirac’s ‘electron sea’ or for its mysterious ‘holes’.

$$\gamma + e^- \rightarrow \gamma + e^-$$

is ‘really’ the same process as pair annihilation

$$e^- + e^+ \rightarrow \gamma + \gamma$$

although in the laboratory they are completely different phenomena.

Mysterious Neutrinos

β decay

$$A \rightarrow B + e^-$$

Conservation of charge requires that B carry one more unit of positive charge than A . (We now realize that the underlying process here is the conversion of a neutron, in A , into a proton, in B ; but remember that in 1930 the neutron had not yet been discovered.) Thus the ‘daughter’ nucleus (B) lies one position farther along on the periodic table. There are many examples of beta decay: potassium goes to calcium (${}^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca}$), copper goes to zinc (${}^{64}_{29}\text{Cu} \rightarrow {}^{64}_{30}\text{Zn}$), tritium goes to helium (${}^3_1\text{H} \rightarrow {}^3_2\text{He}$), and so on.

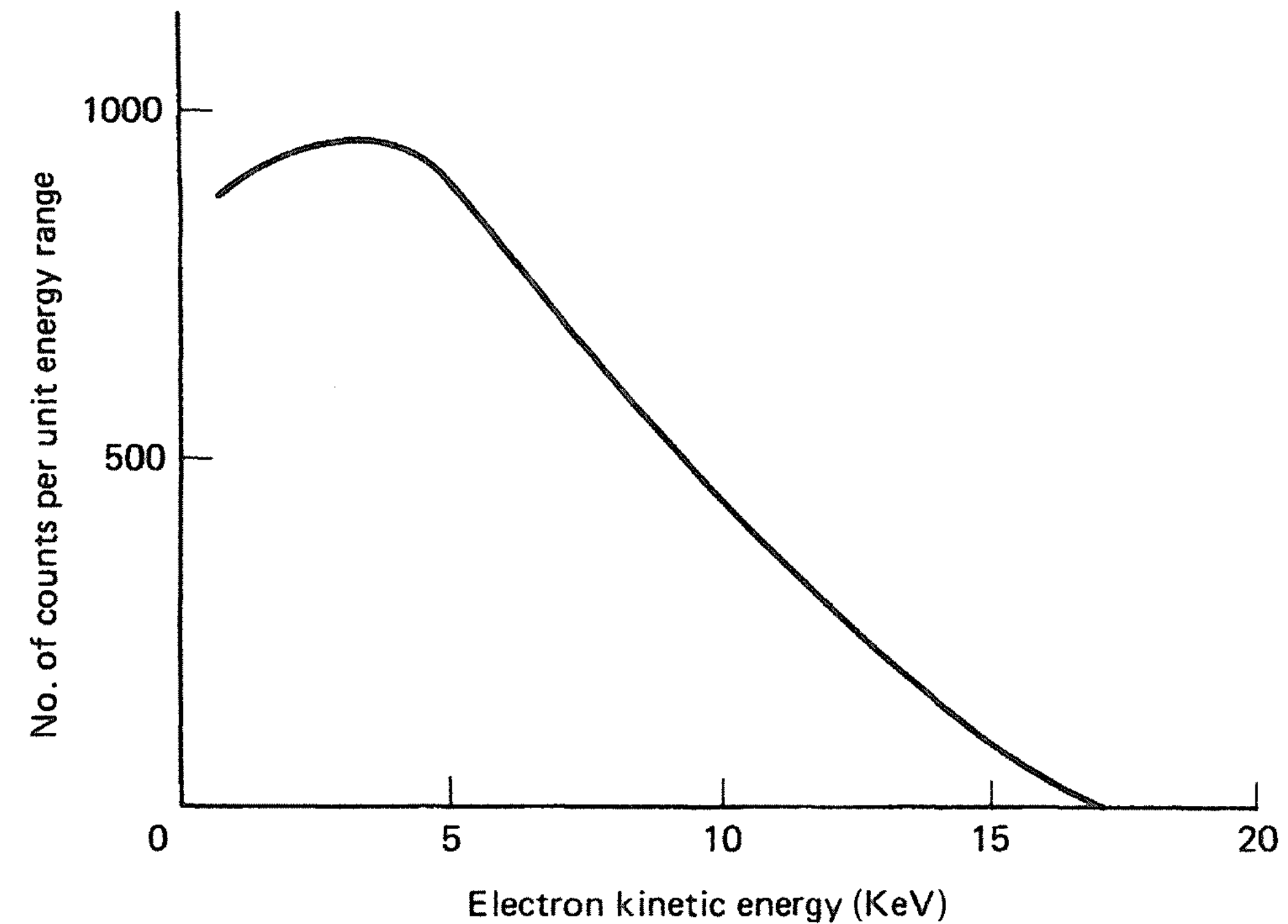
Now, it is a characteristic of two-body decays ($A \rightarrow B + C$) that the outgoing energies are kinematically determined, in the center-of-mass frame. Specifically, if the ‘parent’ nucleus (A) is at rest, so that B and e come out back-to-back with equal and opposite momenta, then conservation of energy dictates that the electron energy is

$$E = \left(\frac{m_A^2 - m_B^2 + m_e^2}{2m_A} \right) c^2$$

In modern terminology, the fundamental beta decay process is

$$n \rightarrow p^+ + e^- + \bar{\nu}$$

(neutron goes to proton plus electron plus antineutrino).



Joy in reading Griffiths:

This was a most disturbing result. Niels Bohr (not for the first time) was ready to abandon the law of conservation of energy.[†] Fortunately, Pauli took a more sober

* The upper number is the *atomic weight* (the number of neutrons plus protons) and the lower number is the *atomic number* (the number of protons).

[†] It is interesting to note that Bohr was an outspoken critic of Einstein’s light quantum (prior to 1924), that he mercilessly denounced Schrödinger’s equation, discouraged Dirac’s work on the relativistic

electron theory (telling him, incorrectly, that Klein and Gordon had already succeeded), opposed Pauli’s introduction of the neutrino, ridiculed Yukawa’s theory of the meson, and disparaged Feynman’s approach to quantum electrodynamics. Great scientists do not always have good judgment – especially when it concerns other people’s work – but Bohr must hold the all-time record.

Mysterious Neutrinos

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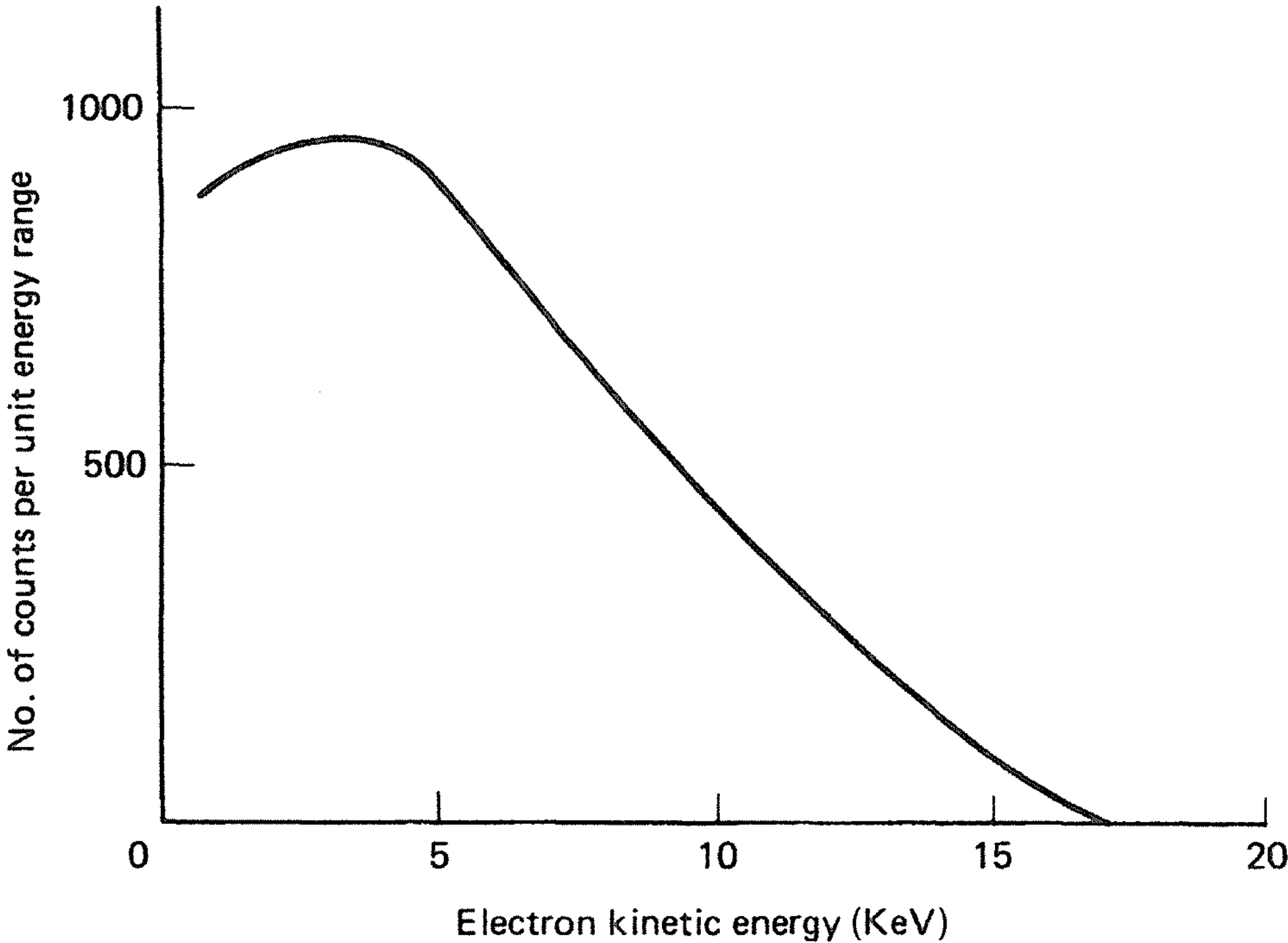


Table 1.1 The lepton family, 1962–1976			
	Lepton number	Electron number	Muon number
Leptons			
e^{-}	1	1	0
ν_e	1	1	0
μ^{-}	1	0	1
ν_{μ}	1	0	1
Antileptons			
e^{+}	−1	−1	0
$\bar{\nu}_e$	−1	−1	0
μ^{+}	−1	0	−1
$\bar{\nu}_{\mu}$	−1	0	−1

Strange Particles

For a brief period in 1947, it was possible to believe that the major problems of elementary particle physics were solved. After a lengthy detour in pursuit of the muon, Yukawa's meson (the π) had finally been apprehended. Dirac's positron had been found, and Pauli's neutrino, although still at large (and, as we have seen, still capable of making mischief), was widely accepted. The role of the muon was something of a puzzle ('Who ordered *that*?' Rabi asked) – it seemed quite unnecessary in the overall scheme of things. On the whole, however, it looked in 1947 as though the job of elementary particle physics was essentially done.

In December 1947, Rochester and Butler discovered a neutral particle decaying to two pions

$$K^0 \rightarrow \pi^+ + \pi^-$$

In 1949, another Keon was discovered

$$K^+ \rightarrow \pi^+ + \pi^+ + \pi^-$$

In due course, many more mesons were discovered – the η , the ϕ , the ω , the ρ 's, and so on.

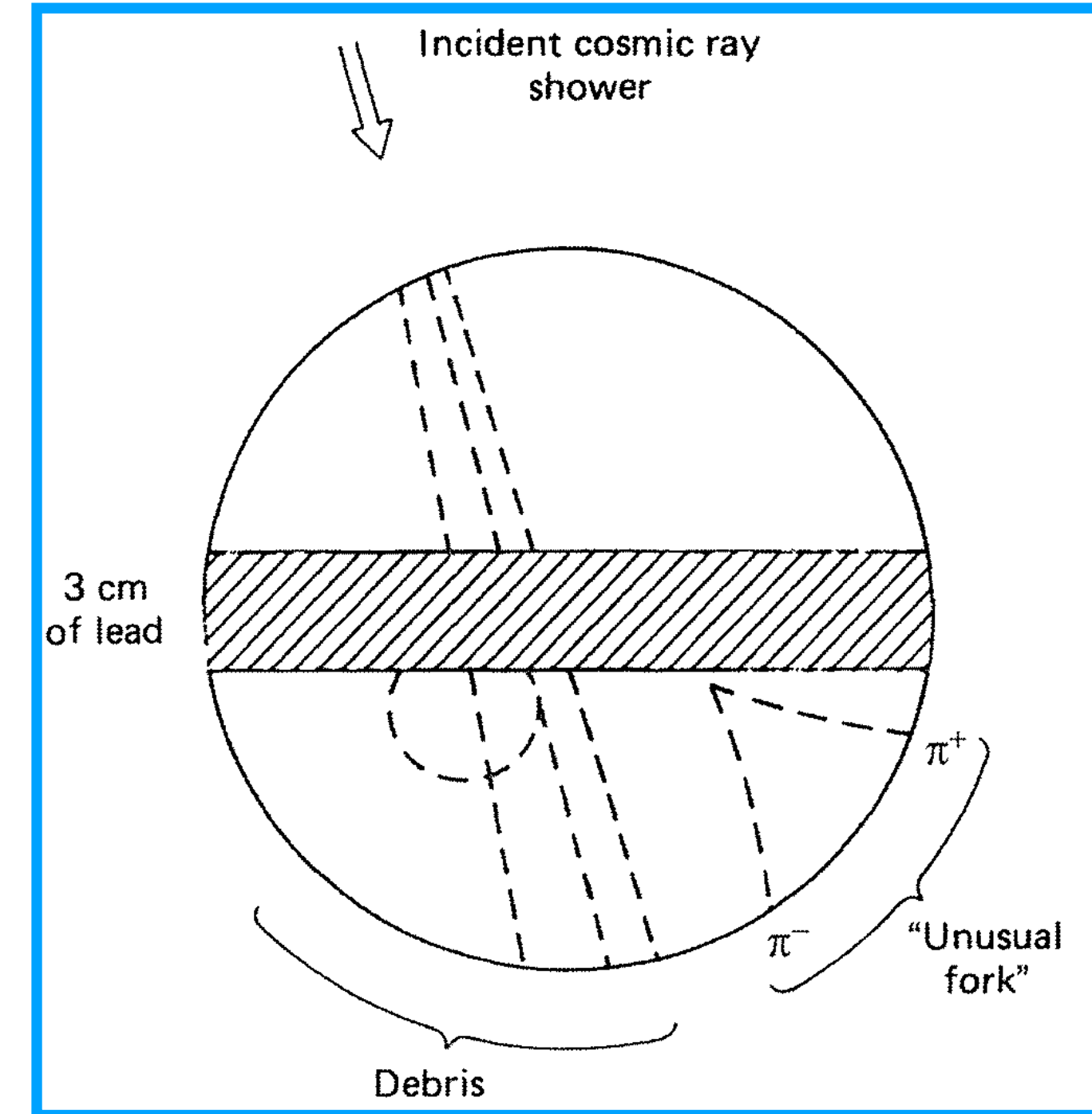
In 1950, another Baryon was discovered

$$\Lambda \rightarrow p^+ + \pi^-$$

'Why is the proton stable?' Why, for example, doesn't it decay into a positron and a photon:

$$p^+ \rightarrow e^+ + \gamma$$

Baryon Number



Strange Particles

It is some measure of the surprise with which these new heavy baryons and mesons were greeted that they came to be known collectively as ‘strange’ particles. In 1952, the first of the modern particle accelerators (the Brookhaven Cosmotron) began operating, and soon it was possible to produce strange particles in the laboratory (before this the only source had been cosmic rays) ... and with this the rate of proliferation increased. Willis Lamb began his Nobel Prize acceptance speech in 1955 with the following words [19]:

When the Nobel Prizes were first awarded in 1901, physicists knew something of just two objects which are now called “elementary particles”: the electron and the proton. A deluge of other “elementary” particles appeared after 1930; neutron, neutrino, μ meson (sic), π meson, heavier mesons, and various hyperons. I have heard it said that “the finder of a new elementary particle used to be rewarded by a Nobel Prize, but such a discovery now ought to be punished by a \$10,000 fine”.

Not only were the new particles unexpected; there is a more technical sense in which they seemed ‘strange’: they are *produced* copiously (on a time scale of about 10^{-23} seconds), but they *decay* relatively slowly (typically about 10^{-10} seconds). This suggested to Pais and others [20] that the mechanism involved in their production is entirely different from that which governs their disintegration. In modern language, the strange particles are *produced* by the *strong* force (the same one that holds the nucleus together), but they *decay* by the *weak* force (the one that accounts for beta decay and all other neutrino processes). The details of Pais’s scheme required that the strange particles be produced in *pairs* (so-called *associated production*). The experimental evidence for this was far from clear at that time, but in 1953 Gell-Mann [21] and Nishijima [22] found a beautifully simple and, as it developed, stunningly successful way to implement and improve Pais’s idea. They assigned to each particle a new property (Gell-Mann called it

‘strangeness’) that (like charge, lepton number, and baryon number) is conserved in any strong interaction, but (unlike those others) is *not* conserved in a weak interaction. In a pion–proton collision, for example, we might produce *two* strange particles:

$$\begin{aligned}\pi^- + p^+ &\rightarrow K^+ + \Sigma^- \\ &\rightarrow K^0 + \Sigma^0 \\ &\rightarrow K^0 + \Lambda\end{aligned}$$

Here, the K ’s carry strangeness $S = +1$, the Σ ’s and the Λ have $S = -1$, and the ‘ordinary’ particles – π , p , and n – have $S = 0$. But we never produce just *one* strange particle:

$$\begin{aligned}\pi^- + p^+ &\not\rightarrow \pi^+ + \Sigma^- \\ &\not\rightarrow \pi^0 + \Lambda \\ &\not\rightarrow K^0 + n\end{aligned}$$

On the other hand, when these particles *decay*, strangeness is *not* conserved:

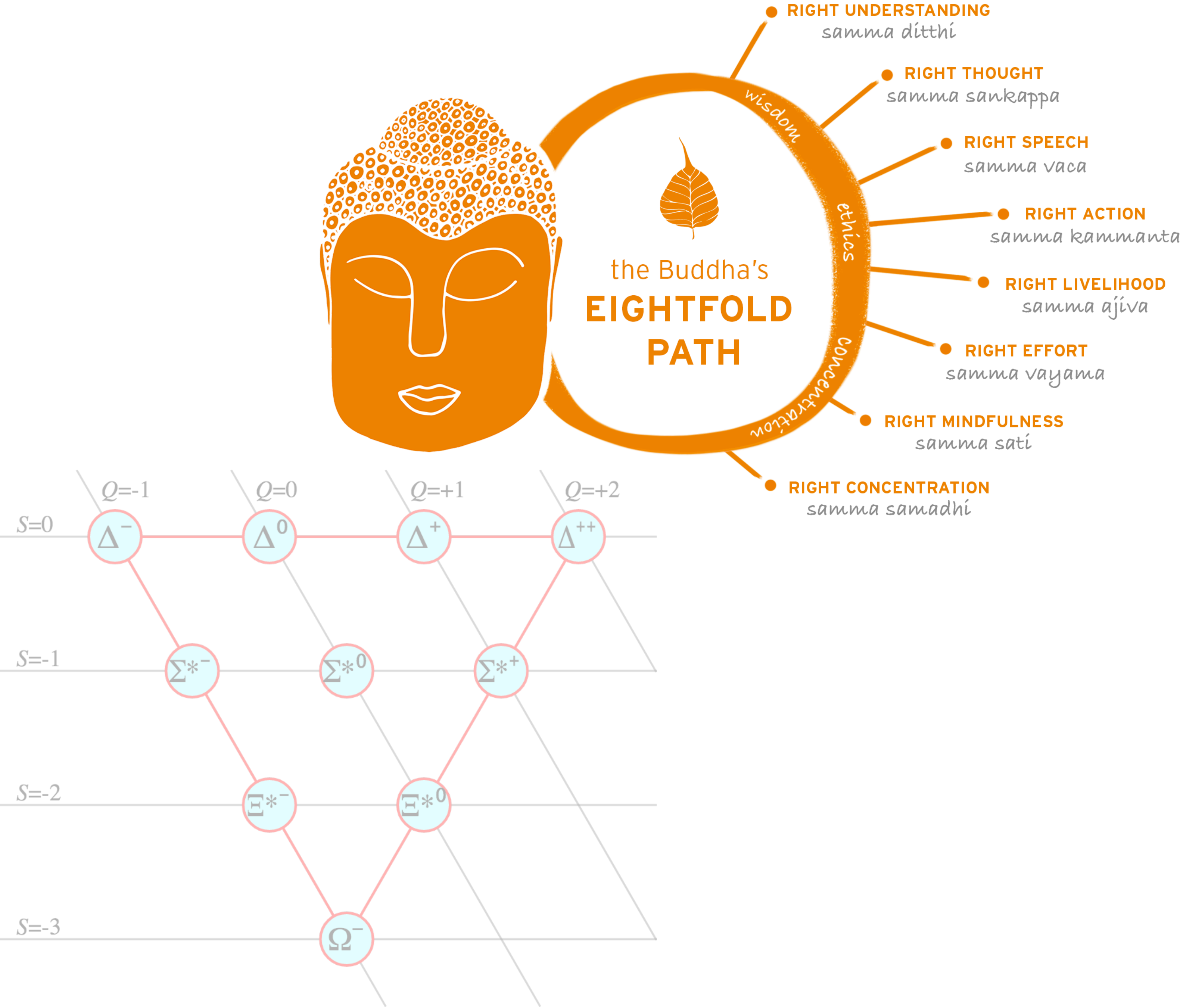
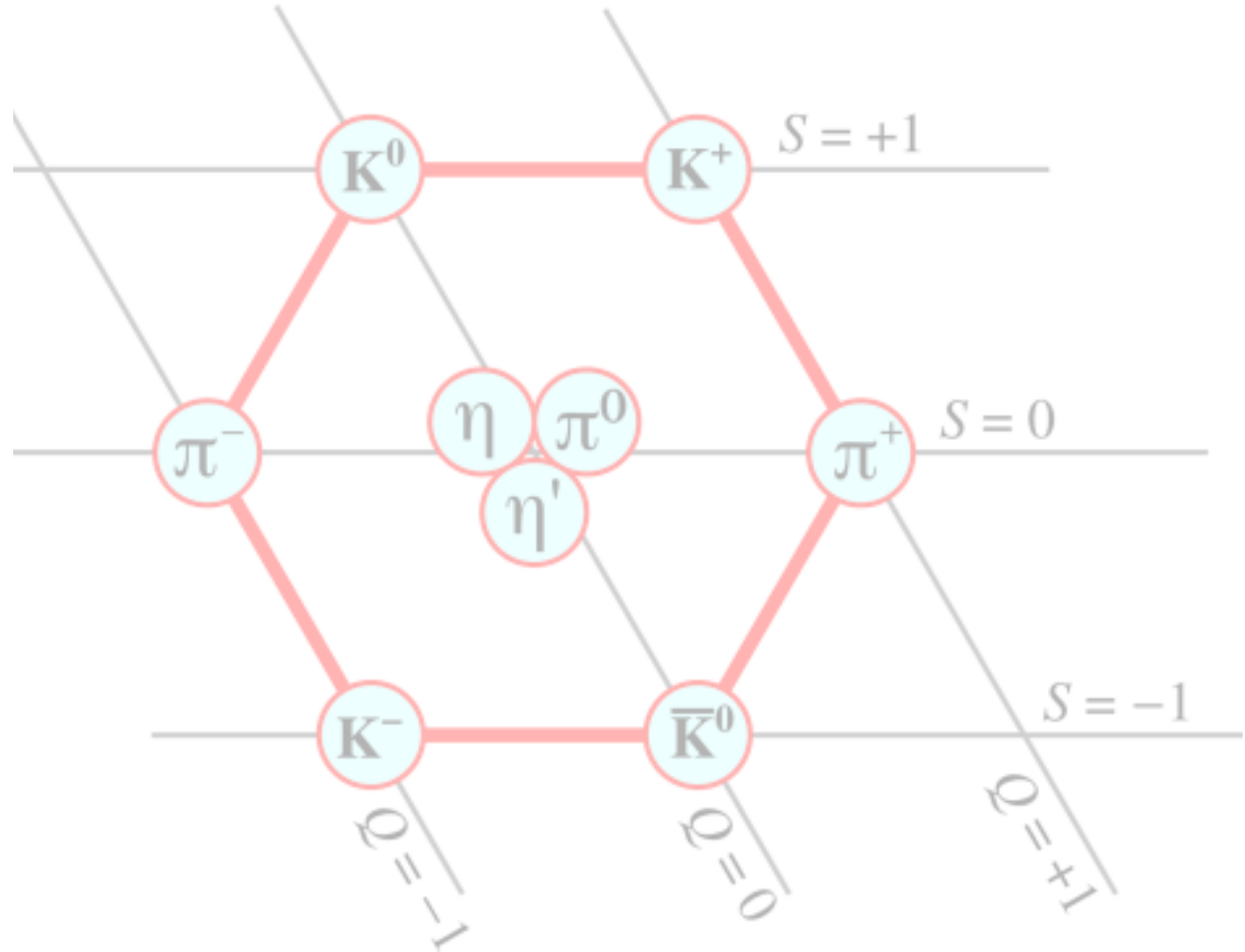
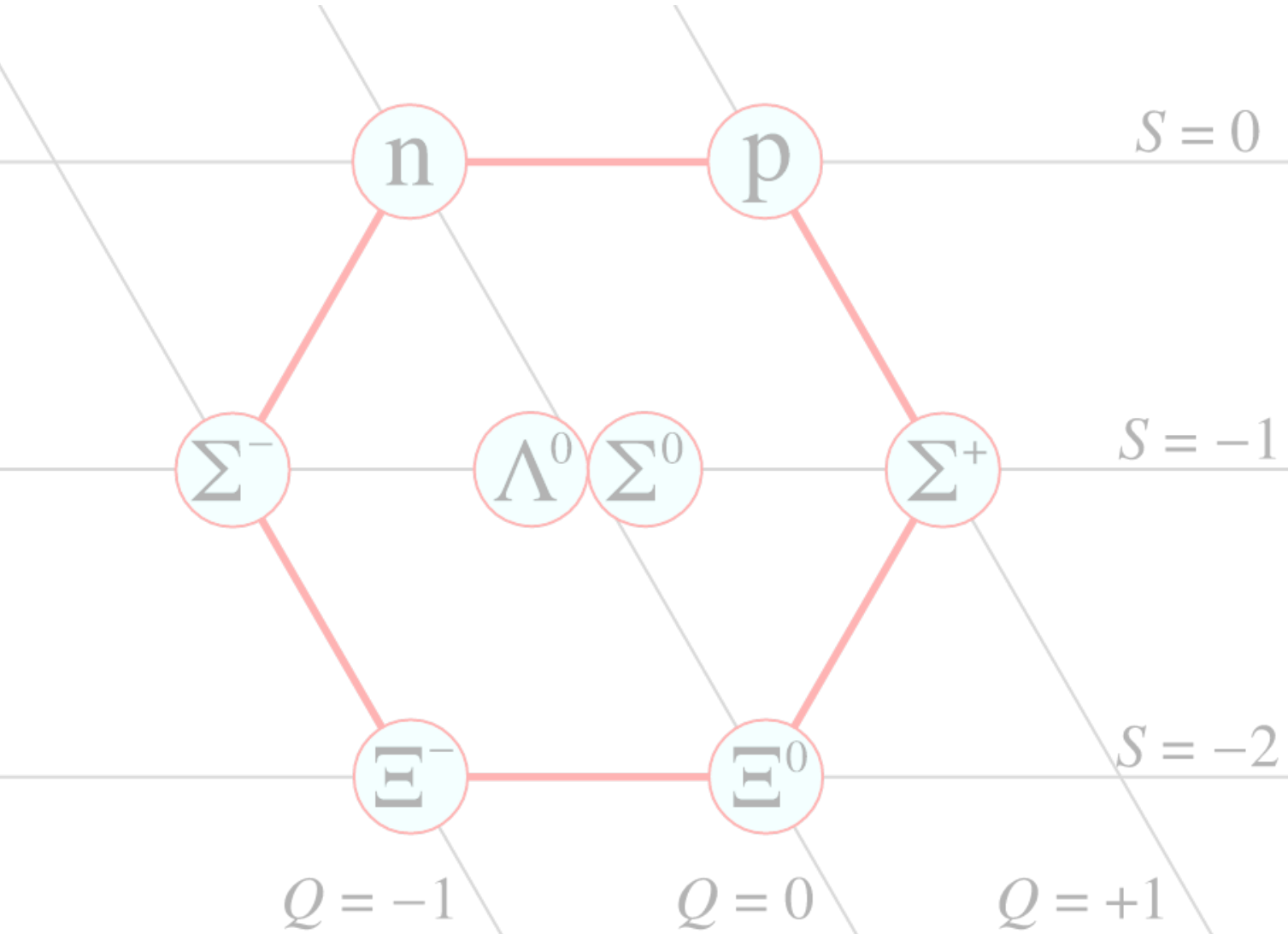
$$\begin{aligned}\Lambda &\rightarrow p^+ + \pi^- \\ \Sigma^+ &\rightarrow p^+ + \pi^0 \\ &\rightarrow n + \pi^+\end{aligned}$$

these are *weak* processes, which do not respect conservation of strangeness.

The Eightfold Way

Things are not random! There is a pattern!

The Eightfold Way



The Eightfold Way

